Computation for Analytics

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Announcement

Last chance to book 1:1 (August 10th)

Class Modality

- Mon (Beginner/Intermediate), Fri (Advanced) + Quiz for the same topics

Assignment 4 (Group HW) - Due by Thursday, August 3re 9 PM

I am not going to share answers/correct answers until one hour after the quiz.

Quiz 4 - Friday, August 4th 1pm - 2pm

- Study list and practice quiz should be available on Canvas

Quiz 5 - Friday, August 11th 10am - 11am

Day 7 - Contents

Numpy

- Installation
- NumPy Array Creation
- Indexing
- Updating Data Types
- Array Manipulation - Transpose, Reshape, Concatenation, Split
- Mathematical Methods
- Custom Functions
- Searching/Sorting/Unique
- Linear Algebra

Day 8 - Contents

Object Oriented Programming

- Class
  - Attributes
  - Methods
- Object
- Class Inheritance

Example 1

Create filtered_rows_sum() which takes file_name and filter rows where the 1st column value is more than twice bigger than the second column value.

- The function should return the sum of both columns in an array.
- Note: You can assume that the file_name always contains two columns and data are only integers separated by comma.
- Ex.

```
filtered_rows_sum(file_name) returns [[83], [89], [79], ...]
```

Example 2

Create concat_columns() which takes file_name and returned integers that are concatenated between the first and second columns.

- Note: You can assume that the file_name always contains two columns and data are only integers separated by comma.
- Ex.

```
concat_columns(file_name) returns [[2961], [19], [5095], ...]
```

Example 3 - 1

Create a class called, School where its constructor takes ***type***, ***teacher_ct***, and ***student_ct.*** For its constructor if its type is 'elementary', 'middle', or 'high', set its ***head*** as ‘principal', whereas set its ***head*** to 'provost' if its type is 'college'.

This class also has a method called ***return_live_with()*** which returns 'parents'.

- Ex.

```
lowell = School('high', 100, 1000)
lowell.return_live_with() returns 'parents'
```

Example 3 - 2

Create a class called College which inherits from School and takes ***teacher_ct*** and ***student_ct*** for its constructor, and set its ***type*** as ‘college’.
Also, override its ***return_live_with()*** to return 'friends'.

- Ex.

```
usf = College(100, 10000)
usf.type == 'college'
usf.head ==  'provost'
usf.return_live_with() == 'friends'
```

NumPy



NumPy Array Creation

1. Converting Python sequences to NumPy arrays with optional dtype (indicating data type)

```
np.array(python_array[, dtype=float])
```

2. Reading from a file

```
np.loadtxt(fname=file_path,
           delimiter=‘,',
           ndim=n1,
           dtype=[(‘field_name', data_types),
                  ('field_name', data_types),...],
           skiprows=n2)
```

ex01-np_array.ipynb

https://numpy.org/doc/stable/user/basics.creation.html
https://numpy.org/doc/stable/reference/arrays.dtypes.html
https://numpy.org/doc/stable/reference/generated/numpy.loadtxt.html#numpy.loadtxt

NumPy



Indexing

Function	Description	Example
`array[row][col] or array[row, col]`	**Single Element Indexing** - returns the data located at the given row and col	`np_array = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])` `np_array[:, 2] # returns array([3, 6, 9])`
`array[start:stop:step] [start:stop:step]`	**Slicing and striding** - returns the data located from start upto stop using step as a stride.	`np_array[0:2:1, 1:4:2] # returns array([[2],[5]])`
`array[array_condition]`	**Boolean Indexing** - returns the data that meets the array_condition.	`np_array[np_array[:, 0] < 5]] # returns array([[1, 2, 3],[4, 5, 6]])`
`array[[row_indices], [col_indices]]`	**Fancy Indexing** - selects a subset of rows in a particular order, you can simply pass a list or ndarray of integers indicating the desired order.	`np_array[[2, 0]] # returns array([3, 7])`

ex02-indexing.ipynb

https://numpy.org/doc/stable/user/basics.indexing.html

NumPy



Updating Data Types

- You can check your NumPy data type by `array.dtype`.
- `array.astype(dtype)` : Create an array by converting array to dtype
  - `dtype`
    - By default, the data-type is inferred from the input data.
    - In addition to Python data types, you can use the following array-protocol type strings.
      - Ex. ‘U20’ means 20-character string
        bootcamp.astype(‘U20') returns
        array(['Dine', 'Shan'], dtype=‘<U20')

Array-protocol type strings	Details
'?'	boolean
'b'/'B'	Signed/unsigned byte
'i'/'u'	Signed/unsigned integer
'f'	floating-point
'U'	Unicode string
'O'	(Python) objects
'M'	datetime

ex03-data_types.ipynb
https://numpy.org/doc/stable/reference/arrays.dtypes.html

NumPy



Array Manipulation

Function	Description	Example
`np.transpose(array, axes=None)`	Returns the transposed array	`np_array = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])` `np.transpose(np_array) # returns array([[1, 4, 7], [2, 5, 8], [3, 6, 9]])`
`np.reshape(array, int/ tuple of int)`	Returns a new shape to an array without changing its data	`np.reshape(np_array, [9, 1]) # returns array([[1], [2], [3], [4], [5], [6], [7], [8], [9]])`
`np.split(array, indices_or_sections, axis)`	Split an array into multiple sub-arrays	`np.split(np_array, [1, 2], 0) # returns [array([[1, 2, 3]]), array([[4, 5, 6]]), array([[7, 8, 9]])]`
`np.concatenate((array_1, array_2), axis)`	Join a sequence of arrays along a new axis.	`col = np.array([[10], [11], [12]])` `np.concatenate((np_array, col), 1) #returns array([[ 1, 2, 3, 10],[4, 5, 6, 11], [7, 8, 9, 12]])`

ex04-array_manipulation.ipynb

https://numpy.org/doc/stable/reference/routines.array-manipulation.html

NumPy



Mathematical Methods

Function	Description	Example

`np.exp/log(array)`	Return the exponent/logarithm value of each element.	`arr_1 = np.array([1, 2, 3])`
		`arr_2 = np.array([-1, 5, 6])`
		`np.exp(arr_1). # returns [ 2.718 7.389  20.086]`

`np.ceil(array)/floor(array)/around(array, decimal)`	Return the ceiling/floor/round of each element.	`np.ceil(arr_1)  # returns [1. 2. 3.]`
`np.min/max/sum/mean/median/std(array)`	Return minimum, maximum, sum, mean, median, standard deviatio value in array	`np.min(arr_1)  # returns 1`
`np.argmin/argmax(array)`	Return indices of minimum/maximum value.	`np.argmin(arr_1)  # returns 0`
`np.cov(array_1, array_2)`	Estimate a covariance matrix. The covariance matrix element Cij is the covariance of array_i and array_j.	`np.cov(arr_1, arr_2)  # returns array([[ 1., 3.5], [ 3.5, 14.333]])`
`np.add/subtract/multiply/divide/maximum/minimum (array_1, array_2)`	Return an array by adding/subtracting/multiplying/dividing/comparing elements in array_1 and array_2.	`np.add(arr_1, arr_2))  # returns [0 7 9]` `np.minimum(arr_1, arr_2)  # returns [-1  2  3]`
`np.histogram(array, bins)`	Compute the histogram of array.	`arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])` `np.histogram(arr, 3)  # returns (array([3, 3, 4]), array([ 1., 4., 7., 10.]))`
`np.percentile(array, q)`	Compute the q-th percentile of array.	`np.percentile(arr, [10, 90])  # returns array([1.9, 9.1])`

https://numpy.org/doc/stable/reference/routines.math.html

ex05-math.ipynb

NumPy



Creating Custom Functions

- **`numpy.vectorize(function)`** can create a function that can be apply to an array(s).
  - Ex.

```
def myfunc(a, b):
      if a > b:
          return a
       else:
          return b

vfunc = np.vectorize(myfunc)
vfunc([1,2,3],[0,3,4]) # returns [1,3,4]
```

ex06-custom_functions.ipynb

https://numpy.org/doc/stable/reference/generated/numpy.vectorize.html

NumPy



Searching/Sorting/Unique

Function	Description	Example
`np.where(condition, x, y)`	Searching : You can use boolean indexing covered in Indexing for only returning values that satisfy the given condition. x will be returned when condition is True, otherwise y.	`nums = np.array(range(10))` `np.where(nums < 5, nums*10, nums) # returns array([0, 10, 20, 30, 40, 5, 6, 7, 8, 9])`

`np.sort(array, axis)`	Return a sorted array by axis.	`array = np.array([[1, 1, 2, 2],`
		`[2, 2, 3, 3],`
		`[2, 2, 1, 1]])`
		`np.sort(array, axis=1) # returns array([[1, 1, 2, 2],`
		`[2, 2, 3, 3],`
		`[1, 1, 2, 2]])`

`np.unique(array)`	Set routine, Returns the sorted unique elements of an array	`np.unique(array) # returns array([1, 2, 3])`

ex07-search_sort_unique.ipynb

https://numpy.org/doc/stable/reference/generated/numpy.where.html

https://numpy.org/doc/stable/reference/generated/numpy.unique.html

NumPy



Linear Algebra

Function	Description	Example

`np.dot(array_1, array_2)`	Matrix multiplication of two arrays.	`array_1 = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])`
		`array_2 = np.array([[1, 2], [3, 4], [5, 6]])`
		`np.dot(array_1, array_2)  # returns array([[ 22, 28], [ 49,  64], [ 76, 100]])`

`np.linalg.det(array)`	Compute the determinant of array.	`np.linalg.det(array_1)  # returns -9.5162e-16`
`np.linalg.eig(array)`	Compute the eigenvalues and eigenvectors of a square matrix	`np.linalg.eig(array_1)  # returns (array([ 1.6117e+01, -1.1168e+00, -3.3843e-16]), array([[-0.23197, -0.7858,  0.4082],[-0.5253, -0.08675, -0.81658],[-0.8187,  0.6123,  0.4082]]))`
`np.linalg.inv(array)`	Compute the inverse of a matrix.	`np.linalg.inv(array_1)  # returns array([[ 3.1525e+15, -6.3050+15,  3.1525e+15] [-6.3050e+15,  1.2610e+16, -6.30504e+15], [ 3.1525e+15, -6.3050e+15,  3.1525e+15]])`
`np.linalg.svd(array)`	Compute the Singular Value Decomposition (SVD).	`np.linalg.svd(array_1)  # returns (array([[-0.2148, 0.8872,  0.4082], [-0.5206,  0.2496, -0.8165, [-0.8263, -0.3879,  0.4082]]), array([1.6848e+01, 1.0684e+00, 3.3348e-16]), array([[-0.4797, -0.5724, -0.6651], [-0.7767, -0.0757,  0.6253], [-0.4082, 0.8165, -0.4082]]))`

ex08-linear_algebra.ipynb

https://numpy.org/doc/stable/reference/routines.linalg.html